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MIL-F-10073(ORD)
3 January 1950

MILITARY SPECIFICATION

FUZE, MECHANICAL TIME AND SUPERQUICK, M502, LOADING, ASSEMBLING, AND PACKING

1. SCOPE AND CLASSIFICATION

1.1 Scope.- This specification covers one type of fuze, time and impact functioning, for use in high-explosive ammunition.

2. APPLICABLE SPECIFICATIONS AND DRAWINGS

2.1 Specifications and drawings.- The following Ordnance Department drawings and all drawings and specifications listed or referred to thereon, of the issue in effect on the date of invitation for bids, form a part of this specification:

73-7-98 - Fuze, Mechanical Time and Superquick M502, Assembly
81-3-30 - Machine, Jolt Testing
81-3-35 - Machine, Jumble Testing
81-3-37 - Fixture, Jumble Testing

(Copies of specifications and drawings should be obtained from the procuring agency or as directed by that agency. Both the title and identifying number or symbol should be stipulated when requesting copies.)

3. REQUIREMENTS

3.1 Materials.- Materials and parts shall be in accordance with applicable specifications and drawings.

3.2 The fuze shall meet all requirements specified in the applicable specifications and on the applicable drawings.

3.3 Jolt and jumble.- The fuze shall withstand the prescribed tests without having any explosive elements to explode; fuze to come apart; fuze to arm partially or fully; inner parts including explosive elements to become broken, deformed, or displaced in such a manner as to make the fuze unsafe to handle or dangerous to fire; fuze to lose more than 20 percent of the torque between lower cap and body; or movement of set line to exceed 0.16 inch off safe "S" position. (See 4.3.4.)

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3.4 Drop test.- The hammer spring and set back pin shall not arm the fuze when the fuze is dropped 6 feet or less, but shall not require a drop of more than 25 feet to arm the fuze (see 4.3.6).

3.5 Waterproofness test.- Fuze shall withstand a vacuum of at least 3 inches of mercury without showing leakage (see 4.3.5).

3.6 Proving ground.-

3.6.1 Premature burst.- No premature burst (see 6.1 and 6.2) shall occur in the proving ground tests.

3.6.2 Impact test (target).- Not more than one fuze shall fail to cause the projectile to burst with high-order detonation when fired against a chipboard target (see 4.3.8.1).

3.6.3 Time test.- Not more than two duds, two off-times, or one dud and one off-time shall occur in the combined results of all phases of the time test. The mean time and mean deviation in the time-to-burst of the fuzes shall be in accordance with table I.

TABLE I. - Time test requirements.

Weapon	Elevation	Fuze setting	Mean time of flight	Mean deviation
	Degrees	Seconds	Seconds	Second
90-MM AA Gun	12	5	4.89 ± 0.13	0.065
90-MM AA Gun	16	15	14.90 ± 0.23	.115
90-MM AA Gun	27	30	30.17 ± 0.38	.190

3.7 Workmanship.- All parts shall be free from chips, dirt, grease, and other foreign material. The cleaning method used shall not be injurious to any of the parts, and the fuze shall not be contaminated by the cleaning agents used.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling.-

4.1.1 Lot.- A lot shall consist of 2,500 fuzes, maximum. A larger quantity of 5,000 fuzes may constitute a lot when authorized by the contracting officer. Two hundred, plus or minus fifty, shall constitute the first lot manufactured on any contract, except when a new contract immediately follows the manufacture of satisfactory material of the same character.

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4.1.2 Sampling procedure.- All samples shall be selected by the inspector. Fifty-five fuzes shall be selected from each lot as specified in the following paragraphs. Sampling of component materials and components is described in paragraphs covering tests of the materials and components.

4.2 Inspection.- Inspection shall be as specified in Specification 50-0-1. Each fuze in the lot shall be set on safe "S".

4.3 Tests.-

4.3.1 Determination of graphite.- The graphite content of the tetryl to be used in the detonator shall be determined on each lot or sub-lot at the time of loading. Approximately 8 ounces of tetryl shall be selected by the inspector. The graphite content shall be analyzed as follows:

4.3.1.1 Procedure.- Transfer a weighed portion of approximately 5 grams of the sample selected as indicated above to a tared filtering crucible and extract with four 10-ml portions of acetone, maintaining each portion in contact with the residue for approximately 5 minutes before applying suction, breaking up any lumps which may persist, and washing down the sides of the crucible with each 10-ml portion of acetone to insure complete removal of the tetryl. Aspirate the residue until the odor of acetone is no longer detectable, and dry at 100° to 105° C. for one hour. Calculate the weight of residue as percent of graphite and/or metallic stearate in the sample.

4.3.1.2 Retest.- If the analysis indicates that the graphite content exceeds that prescribed on the drawing by not more than 0.25 percent, the lot or sub-lot of tetryl may be reblended and resubmitted for test. If in the retest, the analysis indicates that the graphite content still exceeds that prescribed on the drawing or if in the original test the graphite content of the sample was found to exceed that prescribed on the drawing by more than 0.25 percent, the tetryl may be reworked, after which it may be resubmitted as a new lot.

4.3.2 Determination of moisture content of lead azide.- The moisture content of the lead azide shall be determined at intervals as directed by the inspector. At least one such inspection shall be made on the lead azide used in each lot of fuzes. When such determination is made, transfer an accurately weighed sample of approximately 2 grams of lead azide to a tared moisture dish approximately 2.5 inches in diameter. Place the dish and contents in an oven at 95° ± 2°C. for 1-1/2 hours, cool in a desiccator, and weigh. Calculate the loss in weight as the percent of moisture in the lead azide.

4.3.3 Determination of moisture content of black powder.- The moisture content of the black powder shall be determined for each lot or sub-lot at the time of loading. The allowable moisture content and the method of determining the percentage thereof shall be in accordance with specification JAN-P-223.

4.3.4 Jolt and jumble test of fuzes.-

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4.3.4.1 Jolt.- Assemble each of 10 of the sample fuzes to an adapter (see figure 1) and then assemble to jolt testing machine shown on Drawing 81-3-30. Give each fuze 1,750 jolts in each of the three positions.

4.3.4.2 Jumble.- After completing the jolt test, subject each of the fuzes used in the jolt test to a jumble test. Assemble in jumble fixture (see figure 2) and place one at a time in jumble machine shown on Drawing 81-3-35, and give the machine 3,600 complete revolutions at the speed specified on Drawing 81-3-35. Examine the fuzes upon completion of the test.

4.3.4.3 Retest.- A retest using double the number of samples used in the original test may be made if requested by the contractor, provided that in the original test:

a. No explosive elements exploded
b. Not more than one fuze came apart
c. Not more than one fuze partially or fully armed, or had any inner parts including explosive elements broken, deformed, or displaced in such a manner as to make the fuze unsafe to handle or dangerous to fire.

Failure of any sample shall be cause for rejection of the lot.

4.3.5 Waterproofness test.-

4.3.5.1 Procedure.- Five fuzes shall be selected for the waterproofness test. Any fixture or apparatus satisfactory to the inspector may be used (see figure 3). Each fuze shall be completely submerged in water at 25° ± 2°C. and subjected to vacuum (see 3.5).

4.3.5.2 Retest.- A retest using double the number of samples used in original test may be made if requested by the contractor, provided not more than one fuze shows leakage in the original test. Failure of any fuze in the retest shall be cause for rejection of the lot.

4.3.6 Drop test of fuzes.-

4.3.6.1 Procedure.- The same fuzes used on the jolt and jumble tests and two additional sample fuzes marked for identification shall be assembled to an inert booster, M20, and 3-inch AA Shell, M42, filled with inert material to give service weight or assembled in a simulated fixture and dropped base downward in such a manner as to fall freely and to strike squarely on the base of the shell upon a lead plate 3/4 inch thick. The lead plate shall be made of lead-antimony alloy of the following composition: 90 percent by weight of lead and 10 percent by weight of antimony. The plate shall be cast in a confining ring of steel to prevent the free flow of the lead under impact. This lead plate shall be replaced at any time the surface becomes uneven or seriously deformed. The lead plate shall be placed on a steel anvil at least 2 inches thick. Ratio of anvil weight to falling weight (fuze and shell) shall be at least 15:1.

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4.3.6.2 Retest.- A retest using double the number of samples used in the original test may be made if requested by the contractor, provided not more than one fuze fails to meet the requirements of 3.4. Failure of any fuze in the retest shall be cause for rejection.

4.3.7 Spin function test.-

4.3.7.1 Procedure.- The same fuzes that have been subjected to the tests prescribed in 4.3.4 and 4.3.6 shall be set at approximate intervals of 2.5 seconds. Each fuze shall be assembled to a spin chuck (see figure 4) and rotated at 16,200 r.p.m. The timing results of the tests of this group will not be used as a basis for acceptance or rejection of the lot, except that failure of one fuze to function shall be cause for rejection of the lot.

4.3.7.2 Retest.- A retest using double the number of samples used in the original test may be made if requested by the contractor, provided not more than one fuze fails to function. Failure of any fuze in the retest shall be cause for rejection of the lot.

4.3.8 Proving ground test.-

4.3.8.1 Impact test.- Fire 10 fuzes set "S" (Safe) against chipboard target 0.138 inch thick (see 6.3), placed not less than 300 feet from the muzzle of the gun. Fuzes shall be fired with H.E. loaded shell in a 3-inch AA gun, rifled to one turn in 25 calibers, at muzzle velocity of 2,600 feet per second.

4.3.8.2 Safety test.- Fire seven fuzes in the 3-inch AA Gun, 13° elevation, rifled to one turn in 25 calibers, at a muzzle velocity of 2,600 feet per second. The fuzes shall be set at 10-second settings. The results of the tests of this group will not be used as a basis for acceptance or rejection of the lot, except that a premature burst shall be cause for rejection. The test will be used by the Ordnance Department for informational purposes. Experience indicates that 90 percent of the fuzes in this group will function within the following limits:

Mean time of flight, 9.45 ± 0.20 seconds.
Mean deviation, 0.180 second.

4.3.8.3 Time test.- Three groups consisting of seven fuzes each shall be fired in the 90-MM AA Gun, M1, at a muzzle velocity of 2,700 feet per second. All settings (see table I) shall be made by a fuze setter or wrench using a calibrated fuze to obtain angular displacement data by which test fuzes shall be set. Set fuze by turning cap counterclockwise with respect to the body. Graduated readings on the test fuzes will not be considered in making the setting, but shall be recorded for information. Use only fuze chronograph times in computing mean time or mean deviation. Disregard in computation of mean time or mean deviation any two of the allowable duds or off-times (see 3.6.3 and 6.3). If the time-to-burst for any one fuze is lost by the fuze chronograph, use the electric clock time, corrected to fuze chronograph time, to determine whether that fuze should be considered

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an off-time. The correction factor shall be the mean difference between the electric clock time and the fuze chronograph time, determined by considering each of the other individual fuzes in the same group. Subtract the mean difference from the electric clock time of the fuze that was lost to determine its fuze chronograph time.

4.3.8.4 Retest.- A retest using double the number of fuzes used in the original test in each phase retested may be made if requested by the contractor provided that in the original test:

a. No prematures occurred (see 6.1 and 6.2).
b. Not more than three fuzes failed to function in accordance with the applicable provision of 3.6.3.
c. Mean deviation, or the variation in mean time of flight of any group of fuzes in table I, did not exceed twice that permitted by table I.
d. Not more than two fuzes failed to cause the projectile to burst with high-order detonation when fired against the target (see 4.3.8.1).

The retest shall be limited to that phase of the test in which failures occurred. Any failure in the retest above those permitted in original test (see 3.6) shall be cause for rejection of the lot.

5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packaging shall be as specified on the drawing.

5.2 Packing.- Packing shall be as specified on the drawing.

5.3 Labeling and marking.- The labeling and marking for shipment shall be as specified on the drawing.

5.4 Data cards.- Data cards, 5 x 8 inches, made of commercial manila tag board, approximately 200 pounds ream weight, shall be prepared for each lot of accepted fuzes in accordance with the form specified below. Quantities (approximately 50 cards per lot) and distribution of data cards shall be in accordance with instructions issued by the Office, Chief of Ordnance.

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AMMUNITION DATA CARD

Kind_____ Ammunition Lot No._____
AIC Symbol_____ Quantity in Lot_____
Dwg. No.*_____ Spec. No.*_____
Metal Parts Manufactured by_____ Lot No._____
Loaded and Assembled by_____ Date_____
Contract No._____ Order No._____
Packed_____ In a box.

REMARKS:_____

Certified to by_____, Inspector
Print here name of Inspector, District, Arsenal or Loading Plant, and Sub-District where applicable.

*Include amendment and revision.

6. NOTES

6.1 A premature burst is considered to have occurred in the ballistic time test if a burst occurs in the bore of the gun or if the time of flight is less than 2 seconds.

6.2 A premature burst in the impact test is defined as a functioning of the shell in the gun bore or in flight before impact.

6.3 An off-time is a time difference from mean time of a group sufficient to bring the results outside the acceptance limits (see 3.6.3).

6.4 Chipboard.- Gage list No. 1, Department of Commerce Simplified Practice Recommendations R44-36, Boxboard Thickness (see 4.3.8.1).

6.5 The contract should list all material to be furnished by the loading contractor. Ordinarily, the contract should require the loading and assembling contractor to furnish the metal parts for:

Primer Assembly M29, Dwg. 74-2-43
Detonator Assembly M23, Dwg. 73-2-190
Relay Assembly, Dwg. 73-2-144.

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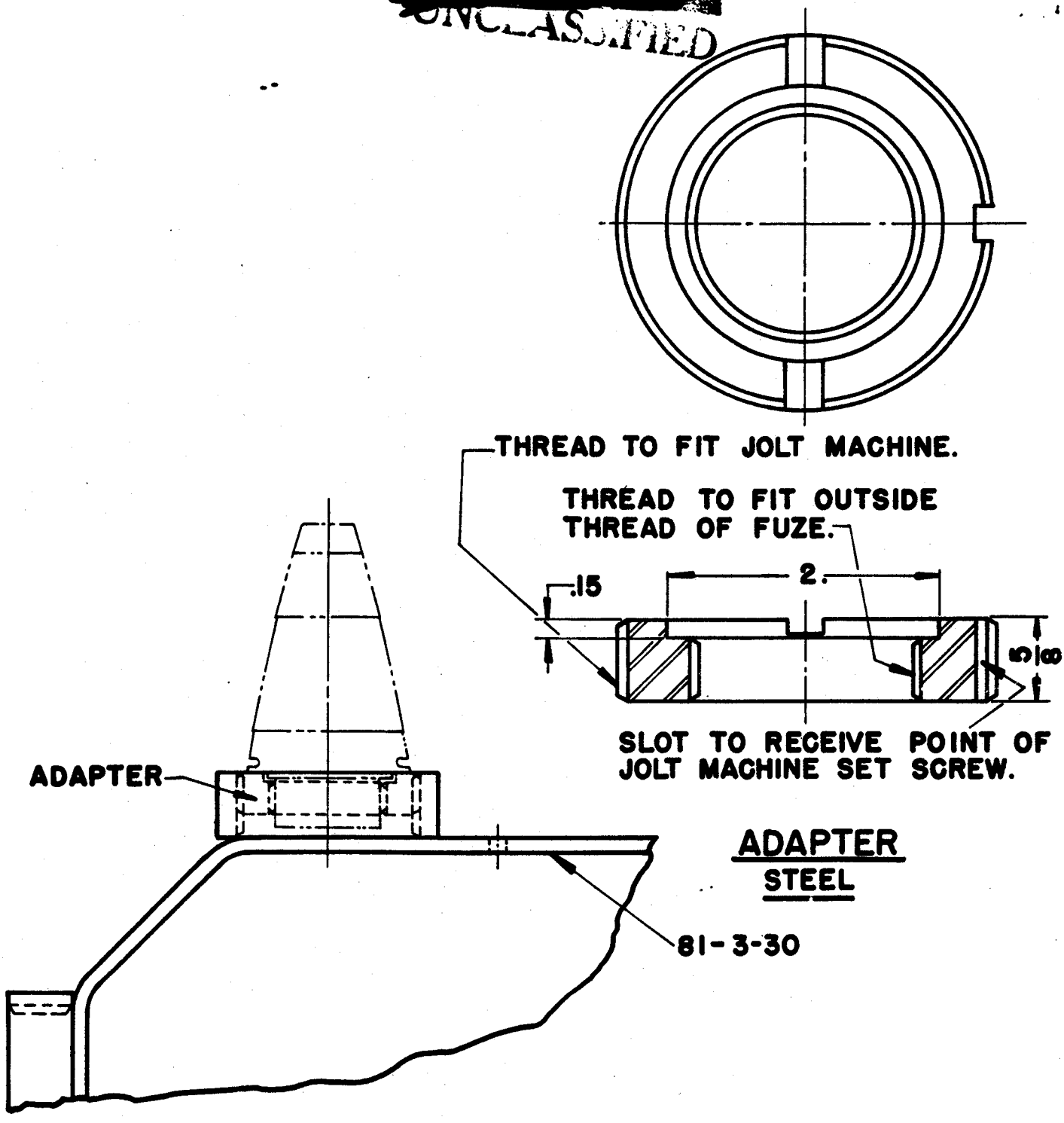
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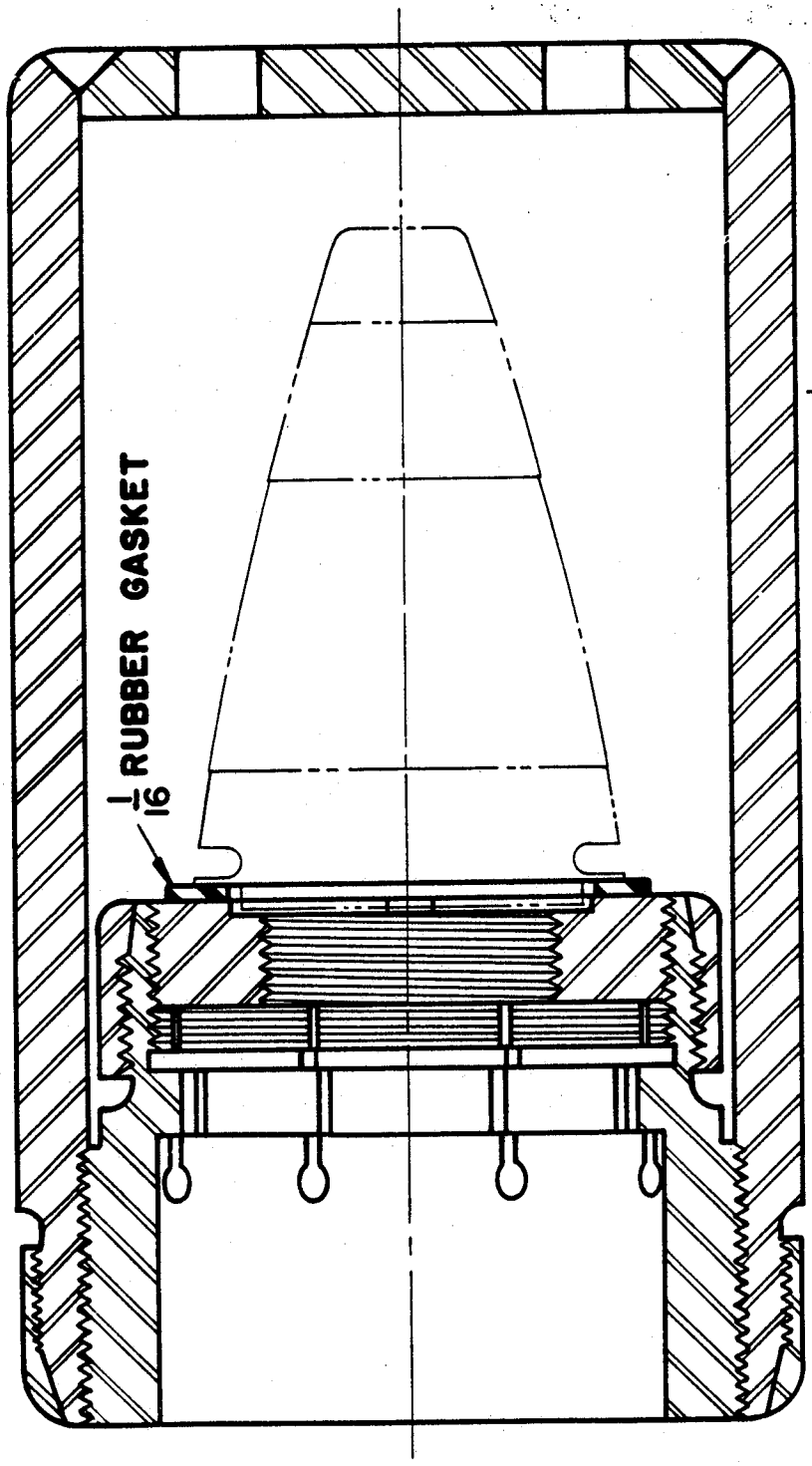


PART VIEW OF JOLT MACHINE

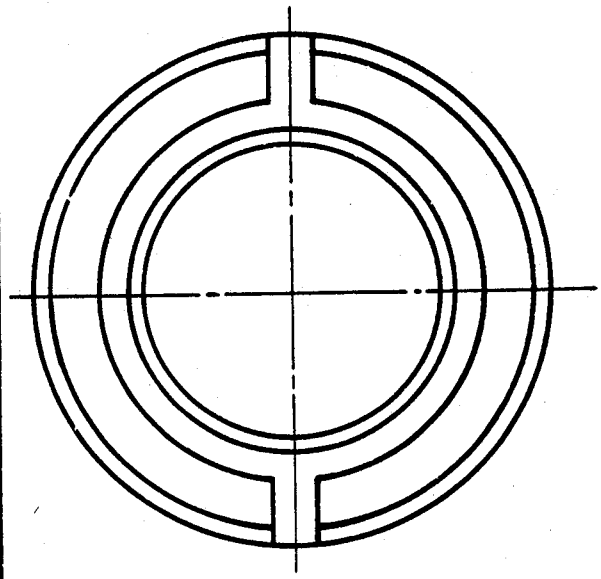
FIG. 1

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JUMBLE FIXTURE



ADAPTER
STEEL

FIG. 2

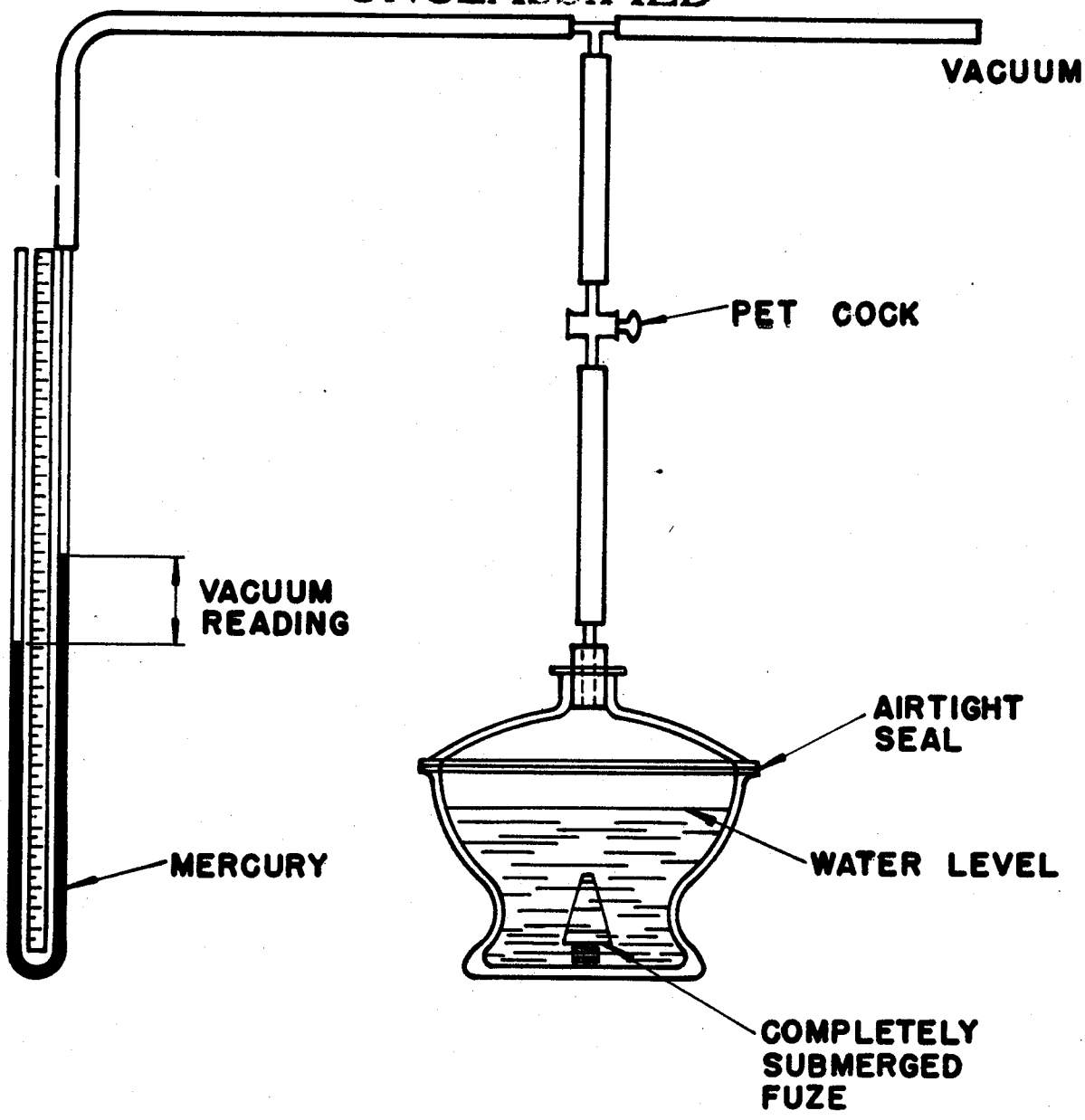
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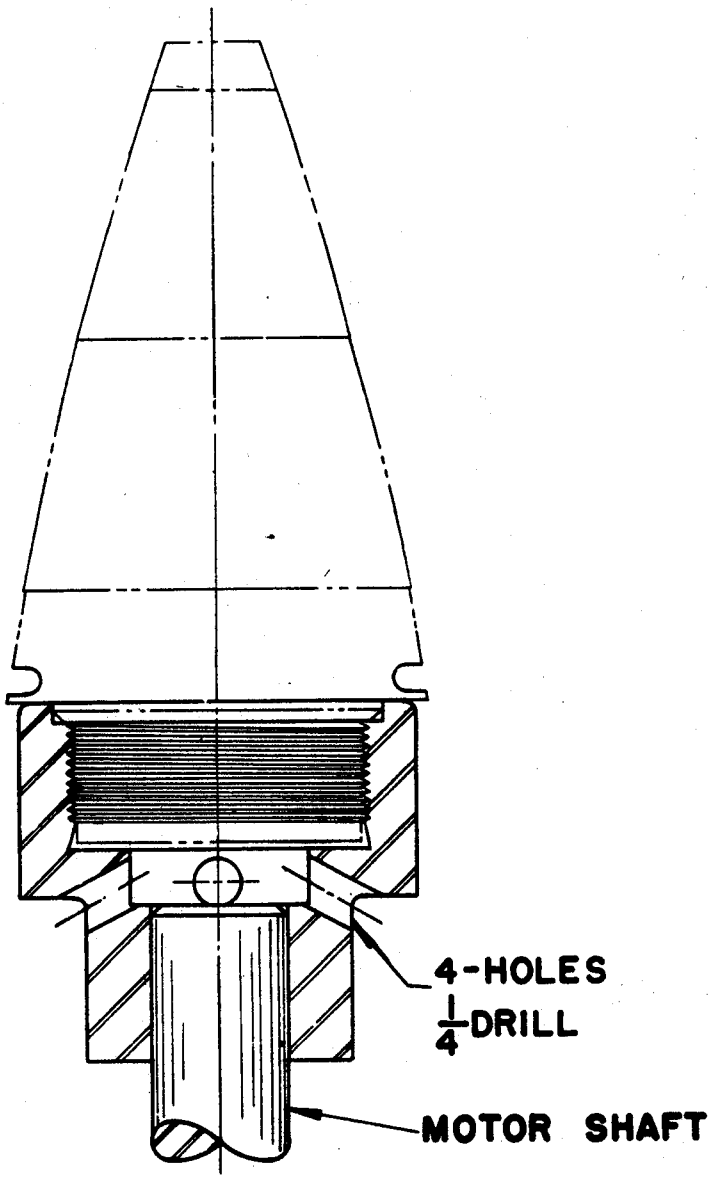
WATERPROOF TEST APPARATUS

FIG. 3

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SPIN CHUCK

TOOL STEEL

HARDEN

HARDNESS ROCKWELL C 50-54

FIG. 4

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